Research on the Performance of different Convolutional Neural Network Models on small Datasets

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Qianjun Yuan ; Zifan Zuo All Authors

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Abstract:
In recent image processing field, as the core of artificial intelligence, the application of machine learning methods has always occupied a very important position. The emergence of convolutional neural networks has undoubtedly played a revolutionary role. As typical models of CNN, resnet, lenet, alexnet models have all achieved great success in the work of processing large image data sets. In this paper, this paper carried out the attribute classification and recognition experiment of 800 Pokémon images using the above three complex models to explore the training effect of complex machine learning models under small data set. Besides, this analyzed the relationship between the training effect of complex models and the size of the dataset.

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SECTION I.
Introduction

Image recognition has important applications in many industries, such as in security systems, lesion identification, traffic congestion detection and other fields [1]. In order to better solve these related problems in the field of computer vision, the concept of CNN came into being. One of the representative methods of deep learning is the convolutional neural network, which is a sort of feed-forward neural network with deep structure that contains convolutional calculations. As convolutional neural networks can define learning and may classify input data in a pan-invariant manner based on their hierarchical structure, they are also known as “Shift-Invariant Artificial Neural Networks.” Besides, CNN is a type of feedforward neural network that contains convolutional computation, which is designed based on the translational invariance of image tasks (objects recognized by images have the same meaning in different locations), thus is good at applying to tasks such as image processing [2],

LeNet, which debuted in 1994, was one of the earliest convolutional neural networks and aided in the development of deep learning as a science. Following multiple successful iterations, Yann LeCun’s revolutionary work has been known as LeNet5 since 1988. According to LeNet5, convolution with learnable parameters is an effective method for extracting comparable features at several locations with a limited number of parameters. At the time, there was no GPU to help with training, and even the CPU was sluggish. As

a consequence, the ability to preserve parameters as well as the calculation process is a huge step forward. The polar opposite of this is using each pixel as a separate input to a huge multilayer neural network. According to LeNet5, because pictures have significant spatial correlations, using independent pixels in the image as separate input characteristics does not take full advantage of these similarities, and such pixels should not be utilized in the first layer. AlexNet networks are based on the paper ImageNet Classification with Deep Convolutional Neural Networks [3], and are a neural network that won the 2010 ImageNet competition. They include image enhancement, Dropout, and other technologies, as well as the ability to distribute the network between two GPUs for computation at the same time, greatly improving computing efficiency. In 2015, ResNet (Residual Neural Network) was proposed by Microsoft Researcher Kaiming He, Xiangyu Zhang etc., by adding residual units to deep neural networks to make the training depth more efficient than before. ResNet was the winner of the 2015 ILSVRC competition, and its topology allows for rapid training of ultradeep neural networks, resulting in a significant increase in model accuracy.

Based on the previous literature, it is obvious that the above three machine learning models have achieved impressive success and breakthroughs in the field of image processing, but it can see that the relevant picture data sets applied to professional research fields are large data sets, while the picture data sets used in many significant research fields tend to have a smaller scale [4], such as Venus surface picture recognition, surveillance image identification of criminal suspects. Therefore, this paper will compare and analyze the training effects of the above three complex models under small-scale datasets to explore their training effects under small-scale data sets.

SECTION II.
Experimental Details and Analytical Methods

A. Data Preprocessing and Argumentation

Pokemon is one of the top IPs around the world, and its popularity has lasted over the last few decades. When the popularity of pokemon comes to the stage of today, there are obviously too many of them, and it may cause confusion when people see a pokemon and cannot tell what kind of pokemon it is. With the help of machine

learning and deep learning, it is reasonable to build a model to at least recognize the main types of most pokemon which are fire, water, and grass.

With the images and data from all generations of pokemon, three different convolutional neural networks were built to predict the primary type of pokemon among water, fire, and grass. The pokemon image dataset contains images of each individual pokemon and a csv file where each column is corresponding to the name of pokemon, the primary type of pokemon and the secondary type of pokemon. While some pokemons only have one type, most of the type two columns will have a null value. Each pokemon has one image in this dataset, and there are in total 809 of them. All images have a blank background, and the focus of the image is in the center which creates a strong contrast.

Because deep learning requires large amounts of data, which can be difficult to get by in some cases, data augmentation is a key aspect of the process. Data augmentation comes to the rescue in these situations. It allows us to increase the amount of the dataset and add variety to it.

Currently, the main approaches to data augmentation consist of two: an unsupervised approach and a supervised approach. Among them, the former adopts preset rules to enrich the data on the existing dataset, including methods such as affine transformation and color transformation. However, this method does not substantially change the data set; the latter uses the distribution followed by the model learning data to randomly generate data consistent with the distribution, which is more conducive to improving the performance of the deep learning model. As an unsupervised learning model, generative adversarial network GANs can essentially change the generated sample correlation brought about by traditional image enhancement. Therefore, this paper uses GAN for data enhancement to improve the performance of the underwater cable detection model [5].

B. Model Structure

The following Fig. 1 to Fig. 3 show the structure of the three models used in this study.

1) Lenet:

As an early CNN model, LeNet has just 2 convolutional layers, 2 average pooling layers and 3 fully connected layers. Because it has a small amount of parameters to be trained and a simple structure, I decided to design an

orthogonal experiment to find out the influence of various factors on this model. We chose four factors for testing, the number of convolutional layers, the size of filters, the number of filters in each convolutional layer, and the dropout in each convolutional layer. From LeNet to ResNet, the accuracy of the model increases with growing depth, so we thought a deeper model will perform better [6], The size of filters decides how many details a filter can get from pictures, but a smaller filter can also ignore some large-scale structures. The number of filters represent how many parameters need to be trained, a lot of filters can make the model more realistic but can also lead to overfitting. And dropout can let the model don't only rely on some of the high weightneuron nodes, it can help all of the neuron nodes on the net work together and build a more general model.

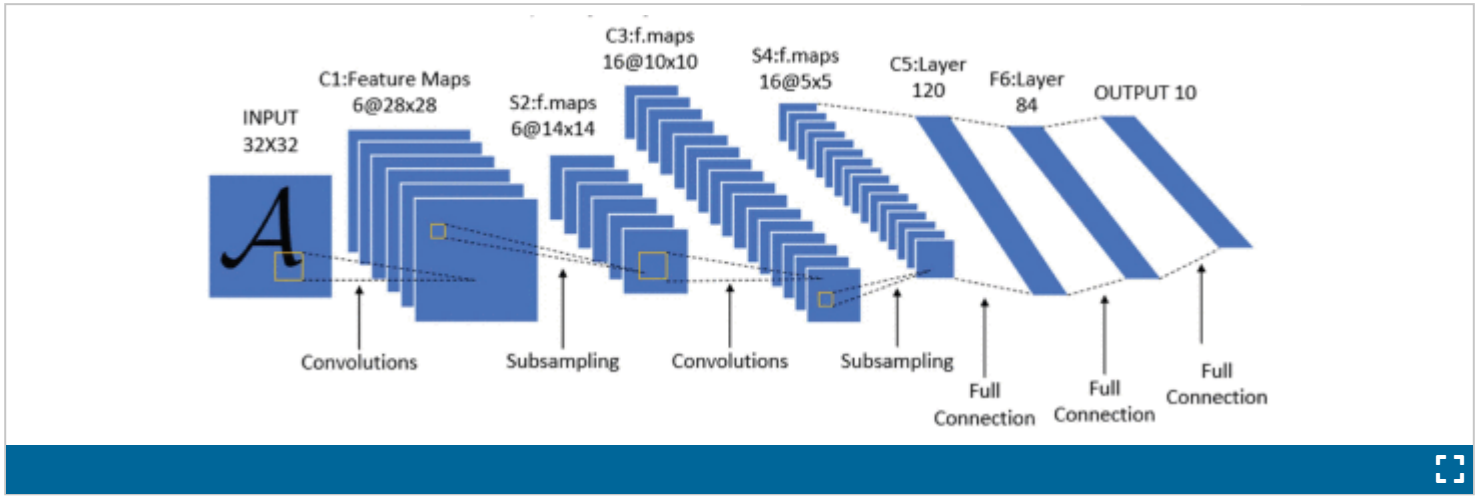

Figure 1.
Structure of LeNet-5

Based on our speculation, we designed an orthogonal experiment with 4 factors and 3 levels. Level 1 is the original structure of LeNet, stride of each convolutional layer is 1, there are a average pooling layers after the first two convolutional layers, the size of fully connected layers in the last of model is 120 - 84 - 3 (the last one is a softmax layer), all of activation function are rectified linear unit (ReLU).

2) Alexnet:

The Convolution neural network architecture used here is AlexNet, and it is one of many methods that produce considerable accuracy. It consists of 5 convolutional layers, 3 max pooling layers, 3 fully connected layers (The last fully connected layer is softmax layer). Above is a visualization of the architecture of AlexNet. For the pre-processing stage, the distribution of pokemon type was analyzed to provide an overall idea of what are some popular pokemon types. Then, all pokemon images paths were added to the dataframe. For this part, only the fire, water, and grass type were for training. For the data augmentation part, since there are no lots of data. Getting more data is necessary while there is no more data. Data augmentation can help by making minor changes to the current pokemon images including rotation, flip, zoom, rescale so that we have more data for the model, and it can also deal with overfitting. Finally, it is part of the AlexNet CNN model. The model optimizer used is adam. The input image has a size of 120 * 120 * 4. The activation used for each convolutional layer is rectified linear unit (ReLU), which introduces nonlinearity in this model, and negative value will be modified to zero, positive value stays unchanged [7], The first and the second convolutional layer are followed by one max pooling layer to size down the parameters and also to keep the feature of images. One flatten layer before the fully connected layers to flatten input image data into one-dimensional array. The first and the second fully connected layer are followed by one drop out layer to randomly discard some neurons to address overfitting in our model. The last softmax layer has avalué of 3 matching the total number of classes.

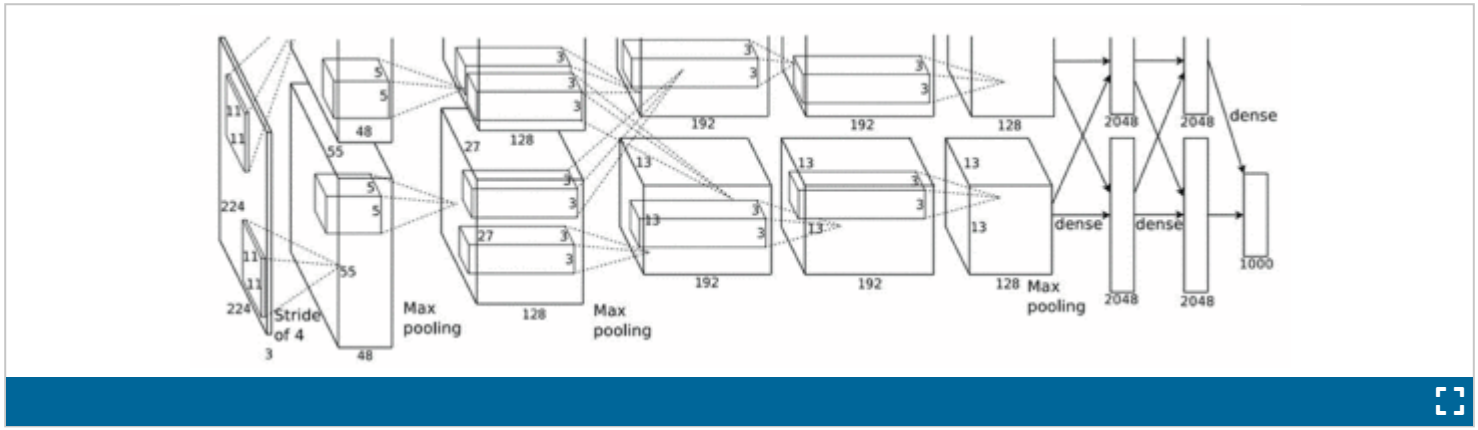

Figure 2
Structure of AlexNet

3) Resnet:

ResNet's main purpose is to provide a channel which is directly linked to the Highway Network. The previous Network layer's output is nonlinearly modified, but the Highway Network layer allows a part of the previous Network layer's output to be maintained. ResNet is similar to the Highway Network in where the original data may be sent to the next level right away. The notion of residual learning was proposed by ResNet. Information loss and loss during information transmission are common problems in convolutional or fully connected networks, and gradient disappearing or gradient expansion will result in deep network training failure. ResNet partially overcomes this problem by skipping the input information on its way to the output, preserving the integrity of the data. The learning objectives and obstacles are simplified since the entire network only has to learn a portion of the variance among outputs and inputs [1].

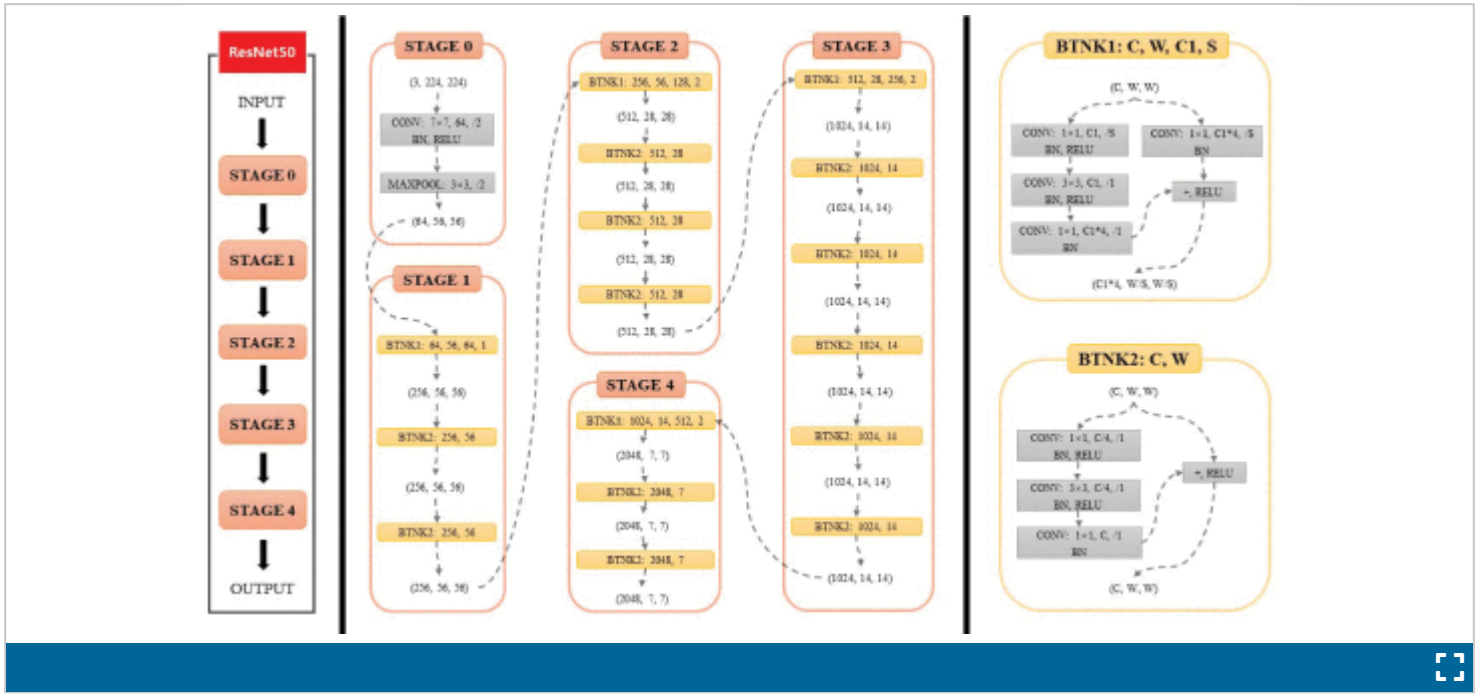

Figure 3.
Structure of ResNet

C. Orthogonal Experiment Design

It can be seen from the development of CNN that the deeper the model is, the better the training performance is. But does this rule still suitable for the small-scale dataset? This study used the same dataset of Pokémon images to conduct comparative experiments on three different CNN models. And to test how factors of a CNN model affect its performance using the Orthogonal Experiment Design (OED) method based on the simplest CNN model, LeNet-5.

The OED method is used to simultaneously study the effect of several factors. It is an efficient and fast experimental design method to select some representative level combinations, based on orthogonality, from the comprehensive experiments and analyze the results to find the optimal level combination. A complete OED is composed of an Orthogonal Array (OA) and the Factor Analysis (FA).

OA is a factorial design array that ensures that any factor's levels are compared equitably. OA is an array of numbers ordered in rows and columns, and each column indicates a distinct component that may be modified from each combination. FA You can quickly estimate the main effects without worrying about results being distorted by individual factor effects also on evaluation function, rating the most key determinants, and selecting the optimal level for each factor using OA's tabulation of experimental results, ensuring that the evaluation function is optimized [8],

In a CNN model, there are many things changeable. For the whole structure, there are three kinds of layers, convolutional layer, pooling layer, and dense fully connected layer, we can change their arrangement and quantity. For each layer, we can change the percentage of dropouts and its activation function. For convolutional layers, we can change the kernel's size, number, and stride of each step.

Due to the core of CNN being the convolutional layer, this study mainly tests the factors around it. The four selected factors are the number of convolutional layers, the size of each filter, the number of filters in each convolutional layer, and the percentage of dropouts in each layer.

Orthogonal experimental design L_9$(4^A 3)$ in Table 1 shows the methodology used in this study. [9], Level 1 represented the structure of LeNet-5. All of the tested combinations keep the 2 by 2 max-pooling layers, three fully connected layers, and Rectified Linear Unit (ReLU) as activation function [10], The bracketed numbers

in the 5th column mean how many filters should be in the extra layers when convolutional layers are more than 2. In addition to OED, this study also conducted a comparative experiment using AlexNet and ResNet.

TABLE I Orthogonal Experiment Design

Levels	Factors:			
	Number of convolutional layers	*Size of filter*	*Percentage of dropouts*	*Numbers of filters*
1	2	5 * 5	0	6 - 16 - (16)
2	3	7 * 7	0.2	24 - 64 - (64)
3	4	11 * 11	0.5	96 - 256 - (256)

SECTION III.
Results

A. OED Experiment results

Using the OED method and the testing factors listed in Table 1, nine sets of testing were performed. Test results are shown in Table 2. The validation accuracy in this study is the average value of the last 20 epochs of 100 epochs because the value at this time has tended to be stable.

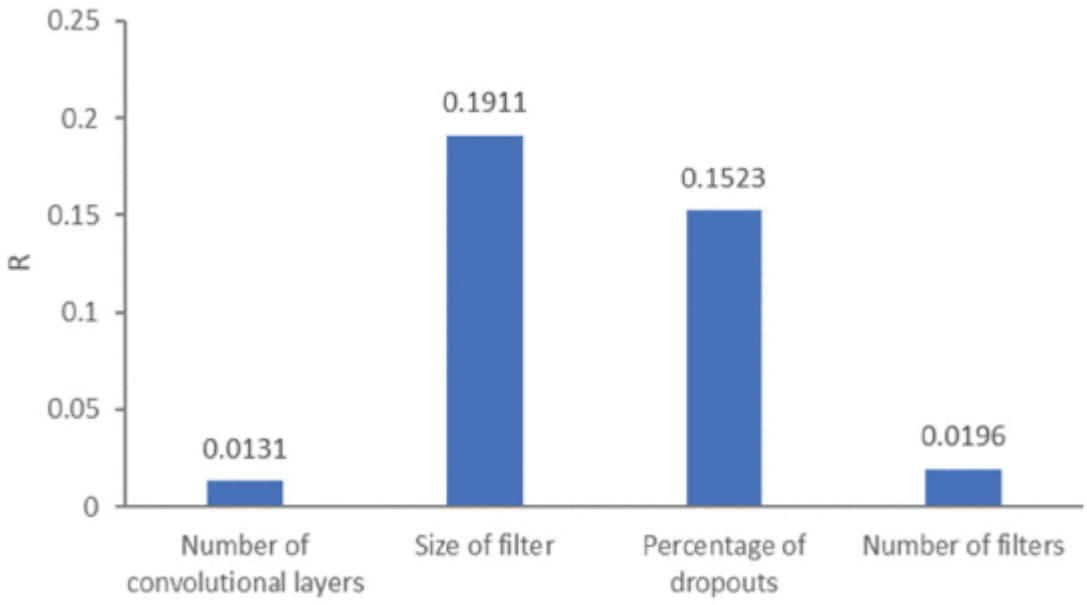

Figure 4.
Range values of different factors

It can be observed from the Intuitive Analysis that, under the conditions of this investigation, the degree of effect of each factor on the validation accuracy is as follows: size of filter > percentage of dropouts > number of filters > number of convolutional layers. The range values of different factors were shown in Fig. 4. The mean values of each factor were shown in Fig. 5 to Fig. 8.

It should be noted that all these charts simply depicted each factor's propensity, not their ability to predict additional values that had not been verified experimentally. The model performs poorly as the filter size and percentage of dropouts increase, according to the change in validation accuracy of the model for each factor. The number of convolutional layers and the number of filters did not show a significant effect in the experiment. The original LeNet-5, which is the combination of A1B1C1D1, achieved the best results. It indicated that the increasement of model complexity in these four aspects cannot improve the model when

handling extremely small datasets. This phenomenon is also evident when the results of the three models LeNet-5, AlexNet, and ResNet are compared.

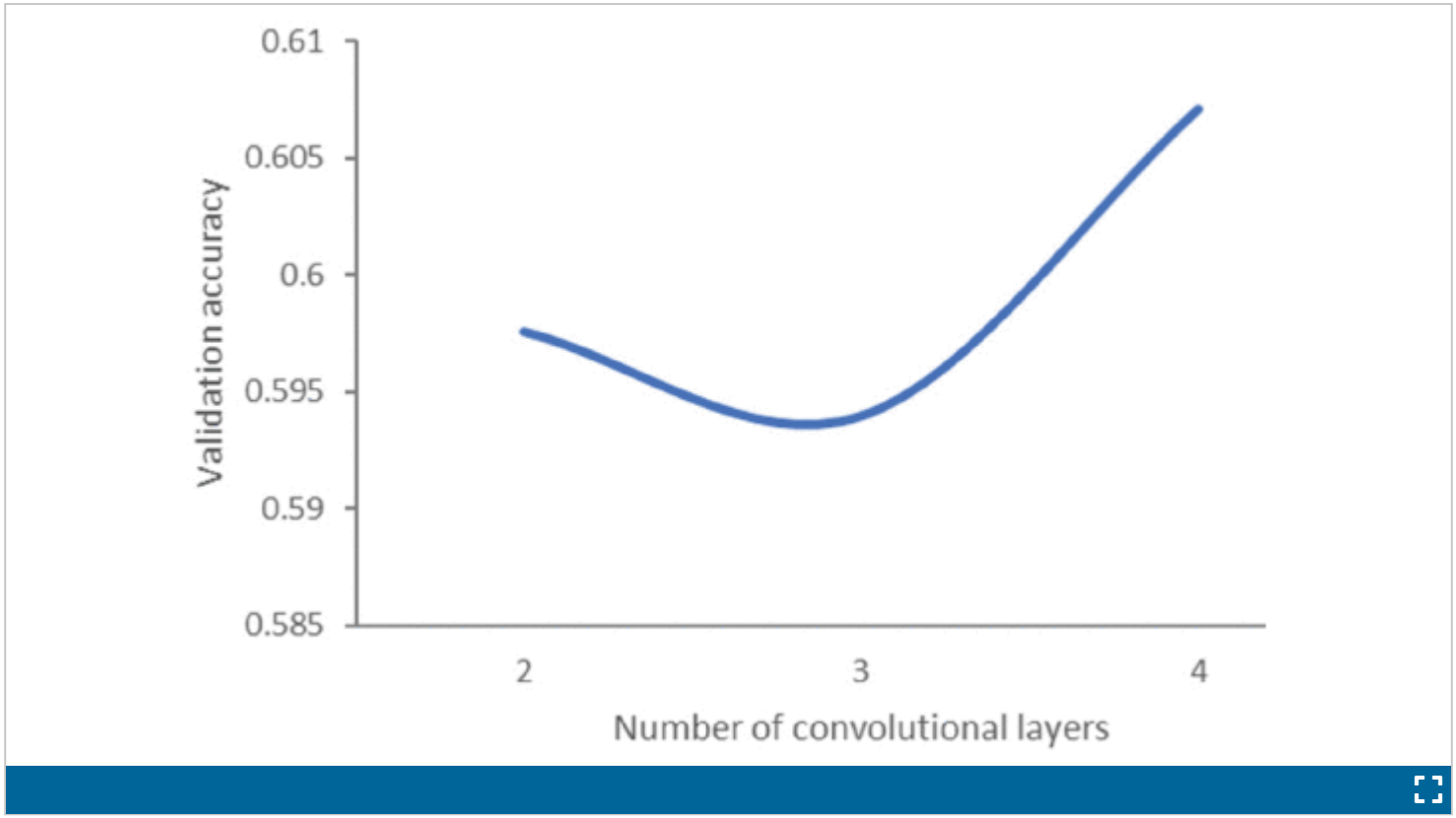

Figure 5
Relationship between number of convolution layers and validation accuracy

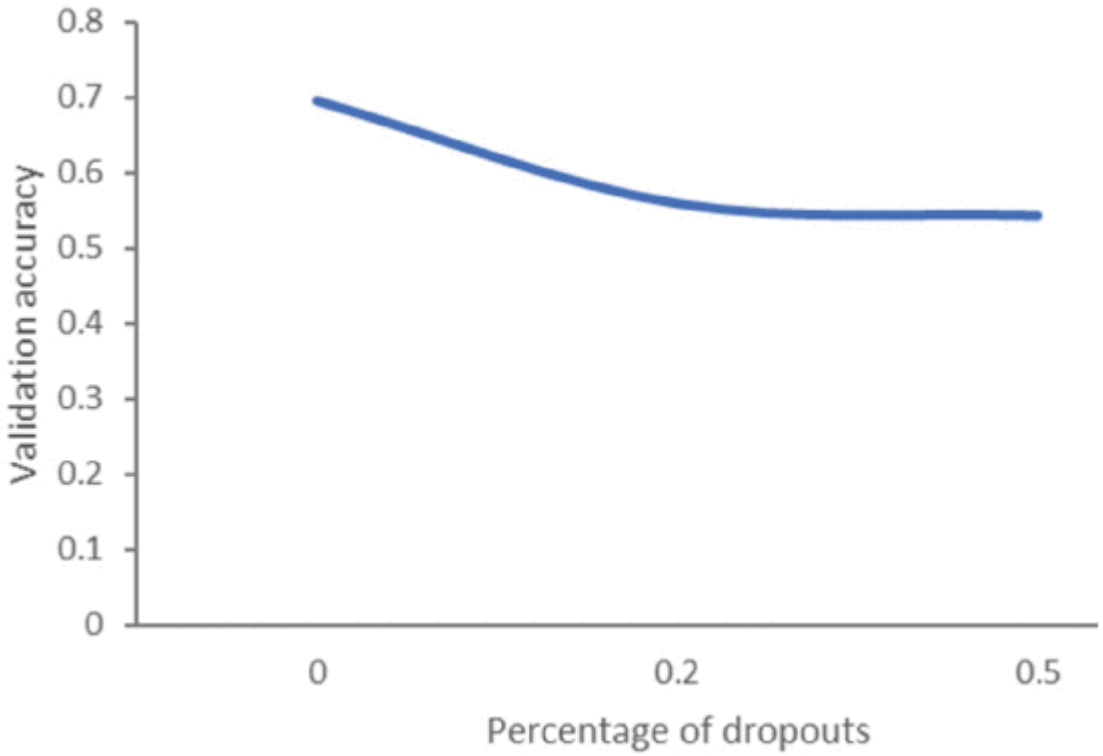

Figure 6
Relationship between size of filter and validation accuracy

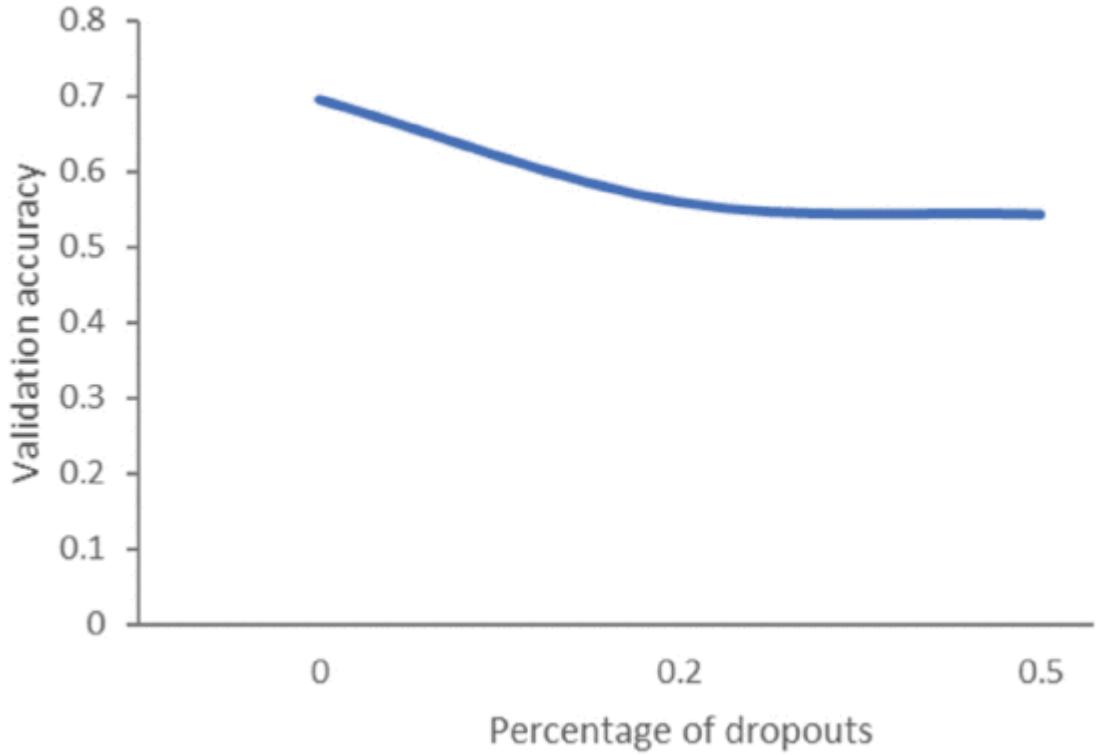

Figure 7.
Relationship between percentage of dropouts and validation accuracy

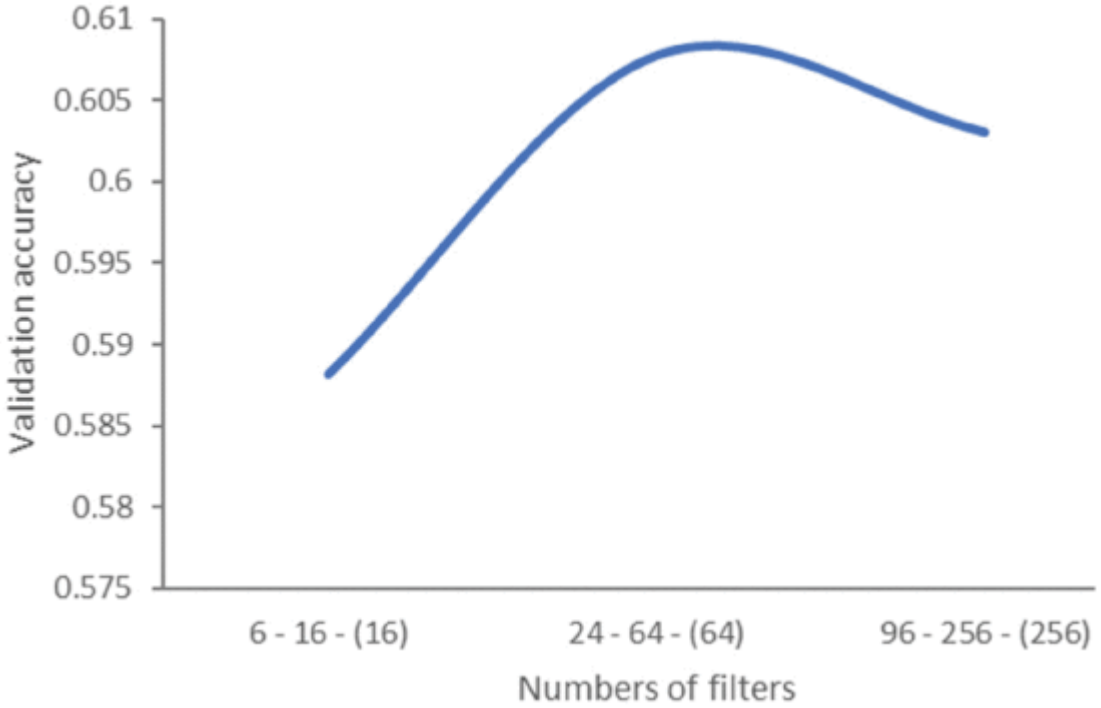

Figure 8
Relationship between size of filter and validation accuracy

B. Intuitive analysis of test results

The test results were subjected to a range analysis, and the findings are provided in Table 3.

TABLE II Results of the Experiment

Testing Number	Factor A	Factor B	Factor C	Factor D	Results
	Number of conv-olutional layers	*Size of filter*	*Percentage of dropouts*	*Numbers of filters*	*Validation accuracy*
1	A1	B1	C1	D1	0.7696
2	A1	B2	C3	D2	0.5661
3	A1	B3	C2	D3	0.4571
4	A2	B1	C3	D3	0.6286
5	A2	B2	C2	D1	0.5589
6	A2	B3	C1	D2	0.5946
7	A3	B1	C2	D2	0.6625
8	A3	B2	C1	D3	0.7232
9	A3	B3	C3	D1	0.4357
AlexNet	-	-	-	-	0.5714
ResNet	-	-	-	-	0.6573

TABLE III Intuitive Analysis of Test Results

Index	A	B	C	D
	Number of conv-olutional layers	*Size of filter*	*Percentage of dropouts*	*Number of filters*
K1	1.7928	2.0607	2.0874	1.7642
K2	1.7821	1.8482	1.6785	1.8232
K3	1.8214	1.4874	1.6304	1.8089
$\overline{K1}$	0.5976	0.6869	0.6958	0.5881
$\overline{K2}$	0.5940	0.6160	0.5595	0.6077
$\overline{K3}$	0.6071	0.4958	0.5435	0.6030
R	0.0131	0.1911	0.1523	0.0196
Ranking	B > C > D > A			
Best combination	A1	B1	C1	D1

Ki represents the sum of results corresponding to each level (i = 1, 2, 3); $\overline{Ki}$ represents the average results for each level (i = 1, 2, 3); R represents $\overline{K_{max}} - \overline{K_{min}}$, more substantial impacts are indicated by a higher R-value.

SECTION IV.
Conclusion

The amount of the data used to feed this model is very limited since no more pokemon could be used in this case, which is not like a daily object such as a coffee mug that has tons of images, even though data augmentation has helped to generate additional images. For normal datasets, the classification accuracy should improve as the model gets deeper. But all of our results are that the deeper the model, the lower the accuracy rate. Apart from the small dataset, this paper thinks it may be that the connection between Pokemon

images and their types is not as strong. There may be other features besides the image that can be used to identity the type of Pokemon. These features are not obvious, and there are few images of each type. It is also possible that there are too many types to be distinguished at the same time, resulting in low accuracy. But most importantly, a complex deep model requires a lot of data to train. Without the support of data, the complexity of the model is meaningless. Therefore, the models tested above are more suitable for large-scale datasets.

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